

HIRADO, Japan

WMO: 478050

Lat: 33.3606N

Lon: 129.5514E

Elev: 59

StdP: 100.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.0	(c) 2.2	(d) -8.0	(e) 1.9	(f) 2.7	(g) -6.7	(h) 2.2	(i) 3.5	(j) 10.2	(k) 3.1	(l) 9.4	(m) 3.5	(n) 5.6	(o) 320	(p) 0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.8	(c) 30.9	(d) 26.0	(e) 30.0	(f) 25.9	(g) 29.1	(h) 25.6	(i) 26.9	(j) 29.2	(k) 26.4	(l) 28.7	(m) 26.0	(n) 28.1	(o) 3.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	28.1	25.8	21.2	27.8	25.4	20.7	27.5	84.9	29.4	83.0	28.8	81.2	28.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 7.9	(c) 7.1	(e) -0.8	(f) 32.6	(g) 1.5	(h) 1.0	(i) -1.9	(j) 33.4	(k) -2.8	(l) 34.0	(m) -3.6	(n) 34.5	(o) -4.7	(p) 35.2		
(a) 8.9	(b) 7.9	(c) 7.1	(e) -3.3	(f) 27.5	(g) 1.1	(h) 0.7	(i) -4.1	(j) 28.0	(k) -4.7	(l) 28.3	(m) -5.4	(n) 28.7	(o) -6.2	(p) 29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	6.9	7.6	10.2	14.3	18.3	21.4	25.5	26.9	23.8	19.4	14.2	9.2
	DBStd	7.19	2.72	2.90	2.75	2.42	1.85	1.72	2.01	1.56	2.09	2.27	2.90	2.97
	HDD10.0	264	102	77	29	2	0	0	0	0	0	0	3	51
	HDD18.3	1473	355	300	251	121	23	1	0	0	0	15	126	282
	CDD10.0	2651	6	11	37	132	258	342	481	525	413	291	129	27
	CDD18.3	819	0	0	0	1	23	93	223	267	163	47	2	0
	CDH23.3	5233	0	0	0	1	24	171	1659	2514	809	56	0	0
	CDH26.7	1074	0	0	0	0	1	4	317	646	105	1	0	0
Wind	WSAvg	3.4	3.7	3.7	3.7	3.5	3.2	3.0	3.4	3.0	3.3	3.3	3.2	3.6
Precipitation	PrecAvg	2067	75	89	135	175	194	321	326	249	205	102	116	81
	PrecMax	2990	216	170	237	388	367	539	608	662	530	245	229	204
	PrecMin	1085	13	22	50	73	59	33	72	22	61	4	9	8
	PrecStd	429	45	43	50	83	82	138	159	154	117	70	61	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	17.4	19.1	22.1	25.2	27.2	31.5	32.6	30.6	26.2	22.2	17.9
		MCWB	13.9	15.0	14.6	16.1	18.5	22.7	26.0	26.0	25.6	22.9	18.3	14.9
	2%	DB	13.6	15.5	17.3	20.7	23.9	25.8	30.3	31.4	29.0	24.7	20.8	16.0
		MCWB	11.0	13.4	14.2	16.2	18.2	22.5	25.9	26.0	24.9	21.2	17.6	12.9
	5%	DB	12.1	13.5	16.0	19.4	22.8	25.0	29.4	30.5	27.8	23.7	19.7	14.8
		MCWB	9.3	10.8	13.3	15.9	18.2	22.6	25.8	26.0	24.4	20.2	16.5	11.6
	10%	DB	11.0	12.0	14.8	18.3	21.8	24.3	28.5	29.8	26.8	22.8	18.7	13.7
		MCWB	8.1	9.1	11.9	15.0	17.9	22.2	25.5	25.8	23.9	19.3	15.6	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.7	16.6	19.4	21.7	24.7	27.5	27.6	26.3	23.8	19.8	16.3
		MCDB	15.5	16.7	17.5	20.5	22.6	25.6	29.5	30.0	28.6	25.5	20.8	17.3
	2%	WB	11.8	13.8	15.0	17.9	20.5	24.1	26.7	27.0	25.7	22.5	18.5	13.8
		MCDB	12.9	15.5	16.6	19.3	21.9	24.9	28.8	29.6	27.9	24.1	20.1	15.2
	5%	WB	9.9	11.5	13.7	16.8	19.7	23.5	26.3	26.6	25.2	21.3	17.3	12.3
		MCDB	11.6	12.9	15.7	18.5	21.4	24.4	28.3	29.1	27.2	23.1	19.2	14.2
	10%	WB	8.4	9.7	12.3	15.7	18.9	22.8	25.8	26.3	24.5	20.0	16.0	11.1
		MCDB	10.6	11.5	14.3	17.5	20.8	23.9	27.8	28.5	26.4	22.0	18.2	13.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	5.6	5.9	5.9	4.5	4.1	4.8	4.5	4.9	5.2	4.8
		MCDBR	6.2	7.0	7.2	7.7	7.6	5.6	5.5	6.0	5.5	5.7	6.2	6.2
		MCWBR	5.2	5.9	5.5	4.7	4.2	3.1	2.4	2.5	2.9	3.6	4.6	5.1
	5% WB	MCDBR	5.9	6.3	6.6	5.8	5.7	3.7	4.7	5.2	4.8	4.9	5.4	5.9
		MCWBR	5.8	6.4	6.2	4.8	4.3	2.7	2.3	2.5	2.8	3.6	4.8	5.6
Clear-Sky Solar Irradiance	taub	0.442	0.482	0.543	0.562	0.569	0.587	0.568	0.537	0.508	0.465	0.437	0.431	
	taud	2.029	1.959	1.819	1.827	1.869	1.918	2.011	2.108	2.125	2.130	2.129	2.068	
	Ebn at Noon	735	750	738	747	747	731	742	760	762	760	739	720	
	Edh at Noon	138	163	203	210	203	193	175	157	148	136	124	126	
All-Sky Solar Radiation	RadAvg	2.13	2.90	4.02	4.86	5.24	4.52	5.12	5.32	4.25	3.56	2.55	1.98	
	RadStd	0.20	0.30	0.28	0.40	0.62	0.56	0.76	0.51	0.42	0.35	0.26	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.30	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page